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| Winter Semester Examination - Dec - 2019                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| Branch: B.Tech. ( Computer Engineering )                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sem: III    |
| Subject with Subject Code: Computer Architecture & Organization[BTCOC304]                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks:60    |
| Date:- 17-12-2019                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Time: 3 Hrs |
| <b>Instructions to the Students:</b><br>1. Each question carries 12 marks.<br>2. Attempt <b>any five</b> questions of the following.<br>3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.<br>4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| Q.1                                                                                                                                                                                                                                                                                                                                               | Solve any following questions.                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| (A)                                                                                                                                                                                                                                                                                                                                               | What, in general terms, is the distinction between computer organization and computer architecture?                                                                                                                                                                                                                                                                                                                                                            | 06          |
| (B)                                                                                                                                                                                                                                                                                                                                               | Explain the computer: the top level structure with structural component with neat sketch diagram.                                                                                                                                                                                                                                                                                                                                                              | 06          |
| Q. 2                                                                                                                                                                                                                                                                                                                                              | Attempt the following questions.                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| (A)                                                                                                                                                                                                                                                                                                                                               | Enlist and explain any two addressing modes. Given the following memory values and a one-address machine with an accumulator, what values do the following instructions load into the accumulator?<br><ul style="list-style-type: none"> <li>Word 20 contains 40.</li> <li>Word 30 contains 50.</li> <li>Word 40 contains 60</li> <li>Word 50 contains 70.</li> </ul> a. LOAD IMMEDIATE 20<br>b. LOAD DIRECT 20<br>c. LOAD INDIRECT 20<br>d. LOAD IMMEDIATE 30 | 06          |
| (B)                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| I.                                                                                                                                                                                                                                                                                                                                                | Convert the following instruction into Accumulator based CPU, Register based CPU.<br>Instruction: $(A*B)-(R+Z)/T$                                                                                                                                                                                                                                                                                                                                              | 03          |
| II.                                                                                                                                                                                                                                                                                                                                               | Is RISC better than CISC? Illustrate your answer with example of processor.                                                                                                                                                                                                                                                                                                                                                                                    | 03          |
| Q.3                                                                                                                                                                                                                                                                                                                                               | Attempt the following questions.                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| (A)                                                                                                                                                                                                                                                                                                                                               | Given $x = 1011$ and $y = 1001$ in two's complement notation (i.e., $x = -5$ , $y = -7$ ), draw and compute the product $p = x * y$ with Booth's algorithm flowchart.                                                                                                                                                                                                                                                                                          | 06          |
| (B)                                                                                                                                                                                                                                                                                                                                               | Show how the following floating-point additions are performed (where significands are                                                                                                                                                                                                                                                                                                                                                                          | 06          |

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|            | truncated to 4 decimal digits). Show the results in normalized form.<br>a. $5.566 \times 10^2 \times 7.777 \times 10^3$<br>b. $3.344 \times 10^1 + 8.877 \times 10^{-2}$<br>c. $6.21 \times 10^5 - 8.877 \times 10^1$                                                             |    |
| <b>Q.4</b> | <b>Attempt the following questions.</b>                                                                                                                                                                                                                                           |    |
| (A)        | What are the differences among direct mapping, associative mapping, and set-associative mapping? A set-associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses. | 06 |
| (B)        | Elaborate the concept of SRAM and DRAM memory with typical memory cell structure.                                                                                                                                                                                                 | 06 |
| <b>Q.5</b> | <b>Attempt the following questions.</b>                                                                                                                                                                                                                                           |    |
| (A)        | What is the overall function of a processor's control unit? A stack is implemented. show the sequence of micro-operations for<br>a. popping<br>b. pushing the stack<br>PUSH 10<br>PUSH 70<br>PUSH 8<br>ADD<br>PUSH 20<br>SUB<br>MUL                                               | 06 |
| (B)        | What is the difference between a hardwired implementation and a microprogrammed implementation of a control unit? <a href="https://www.batuonline.com">https://www.batuonline.com</a>                                                                                             | 06 |
| <b>Q.6</b> | <b>Attempt any two questions.</b>                                                                                                                                                                                                                                                 |    |
| (A)        | In virtually all systems that include DMA modules, DMA access to main memory is given priority than CPU access to main memory. Why?                                                                                                                                               | 06 |
| (B)        | What is the meaning of each of the four states in the MESI protocol? Can you foresee any problem with the write-once cache approach on bus-based multiprocessors? If so, suggest a solution.                                                                                      | 06 |
| (c)        | How does instruction pipelining enhance system performance? Elaborate your answer using RISC instruction stages.                                                                                                                                                                  | 06 |

\*\*\*\*\*End of Paper\*\*\*\*\*